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Calcium Phosphate Emulsions: Possible Applications

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ABSTRACT
A hydrophobic liquid was mixed with a calcium phosphate cement paste and an emulsifier. After intensive mixing, an emulsion was formed. The emulsion was either a W/O emulsion (cement droplets dispersed in oil) or a O/W emulsion (oil droplets dispersed in the cement paste), depending on the emulsifier type and the oil stability. The stability of the two types of emulsion could be well predicted using the hydrophilic-lipophilic balance concept. After the cement had set and the oil had been washed off, calcium phosphate granules or porous blocks were obtained. The composition of the granules and the blocks was either calcium-deficient hydroxyapatite (before sintering) or β-tricalcium phosphate (after sintering). An increase of the emulsifier concentration or a decrease of the oil amount led to a decrease of the block pore size.

INTRODUCTION
Calcium phosphate cements (CPC) were discovered in the 1980's by Brown and Chow [1]. Since then, research has been focused mainly on finding new formulations, understanding their chemistry and biological behavior, and improving their handability. However, very little has been done to exploit the intrinsic property of CPC which is to transform calcium phosphate powders into calcium phosphate blocks at room temperature. In the past, this transformation could only be achieved by means of a sintering process at high temperature. As a result, it is now possible to manufacture a calcium phosphate block with almost any kind of composition, in particular low-temperature calcium phosphates such as dicalcium phosphate dihydrate (DCPD), octocalcium phosphate (OCP) and precipitated hydroxyapatite (PHA). This opens up new possibilities in the medical field. However, for most applications, the calcium phosphate blocks must be porous and have a defined porosity, pore size and pore interconnectivity. Therefore, a suitable manufacturing technique is required. Here, a new technique that could prove very useful is presented. It is based on mixtures of calcium phosphate cement pastes (W) and oils (O), mixtures also called calcium phosphate emulsions (CPE) [2]. The goal of this study was to investigate the effects of several parameters such as the oil amount, the emulsifier type, and the emulsifier amount on the solids obtained with CPE.

MATERIALS AND METHODS
The CPC was made of 8g α-tricalcium phosphate (α-TCP), 2g calcium-deficient hydroxyapatite and 6.0g of an aqueous solution. The solution contained 1% polyacrylic acid 5.1kDa, 0.02% w/w sodium azide, and 0 to 3.5 mg/mL polyethoxylated castor oil (PCO; emulsifier). The oil phase (O) was a mixture of highly viscous paraffin oil and sorbitan monooleate (SM; 0.00, 0.57 and 2.22 mg/ml). Only one emulsifier (PCO or SM) was used at a time. The cement paste was mechanically mixed with 2 to 100g oil phase, i.e. 0.2 to 10 times the volume of the aqueous phase. Initially, the mixture looked inhomogeneous, i.e. the paste had small gray (oil) and white patches (cement). After a certain time of mixing called emulsion time (ET), the paste became macroscopically

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homogeneous, i.e. entirely white. This transition was only observed in O/W emulsions. In W/O emulsions, the paste remained a mixture of gray and white patches. Ten seconds after obtaining an O/W emulsion (after 90s mixing for W/O emulsions), the paste was poured into a syringe whose end had been cut off. When the cement was set, the cement block was pushed out of the syringe, washed, dried in air, and sintered at 1250°C for 4 hours. The blocks obtained by O/W emulsions were broken in the middle, polished and the average pore size was determined by light microscopy. Finally, the crystalline composition of a few unsintered and sintered samples was determined by x-ray diffraction (XRD).

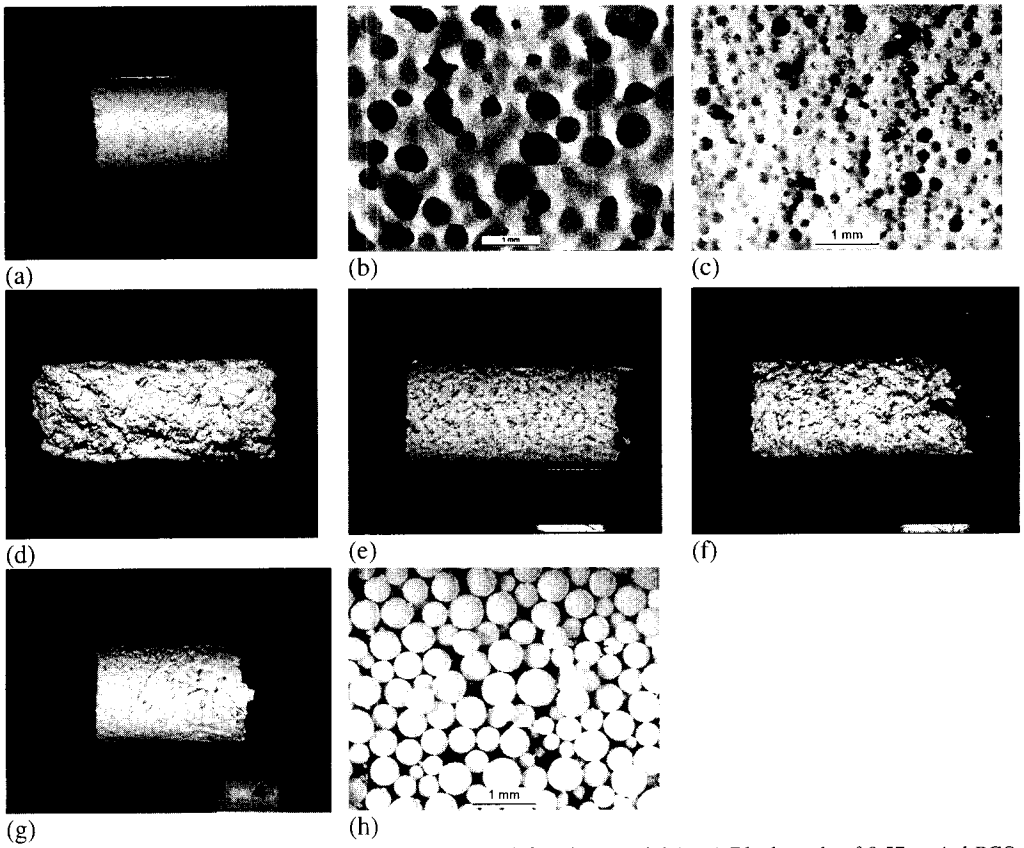


Figure 1: Macrostructures obtained using CPE (from top left to bottom right): (a) Block made of 0.57 mg/ml PCO and 9.6g paraffin oil (54% oil fraction). (b) Same as (a). (c) Block made of 0.57 mg/ml PCO and 7.3g paraffin oil (47% oil fraction). (d) Block made of 0.57 mg/ml PCO and 12.5g paraffin oil (60%). (e) Block made of 0.57 mg/ml SM and 10.0g paraffin oil (54%). (f) Block made of 2.22 mg/ml SM and 10.0g paraffin oil (54%). (g) Block made of 2.22 mg/ml SM and 4.0g paraffin oil (32%). (h) Granules made of 0.57 mg/ml SM and 100g paraffin oil (8%).

RESULTS

The unsintered samples consisted in hydroxyapatite, as evidenced by XRD. No other peaks were detected. After sintering, the samples contained only β-tricalcium phosphate (β-TCP). Therefore, the hydroxyapatite present in the unsintered samples was a calcium-deficient hydroxyapatite (CDHA) with a Ca/P molar ratio close to 1.50.

An O/W emulsion was obtained when the CPC was mixed with a low to moderate amount of paraffin oil and PCO (**Fig 1a**). The oil droplets were dispersed in the CPC paste and the surface of the blocks was smooth. The pores were round and rather homogeneous in size (**Fig 1b, Fig 1c**). Some interconnections were seen between pores, especially at large oil fraction. The pore size

decreased with a decrease of the oil fraction. Above a certain oil fraction, no more O/W emulsion could be obtained. The paste transformed into a W/O emulsion. The resulting block had a rough surface (**Fig 1d**). The block consisted in calcium phosphate lumps and droplets linked by small bridges. The structure was totally open-porous, as evidenced by the rapid loss of paraffine during washing. When PCO was replaced by SM, the transition O/W to W/O emulsion (phase inversion) occurred at much lower oil fraction (compare **Fig 1a** and **1e**). The addition of larger amounts of emulsifier led to more irregular structures (compare **Fig 1e** and **1f**) and shifted the phase inversion to lower oil fractions (compare **Fig 1a** and **1g**). By using a larger amount of paraffin, granules were obtained (**Fig 1h**).

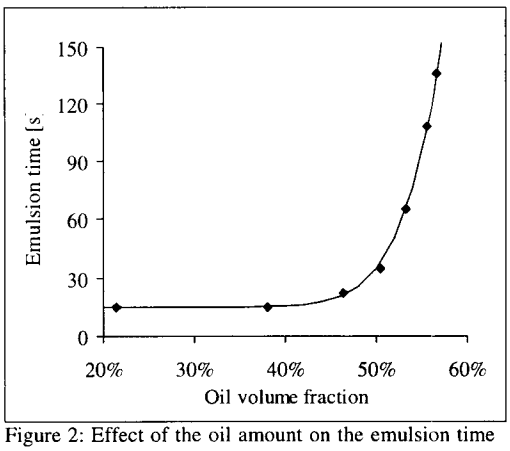


Figure 2: Effect of the oil amount on the emulsion time of the O/W emulsion.

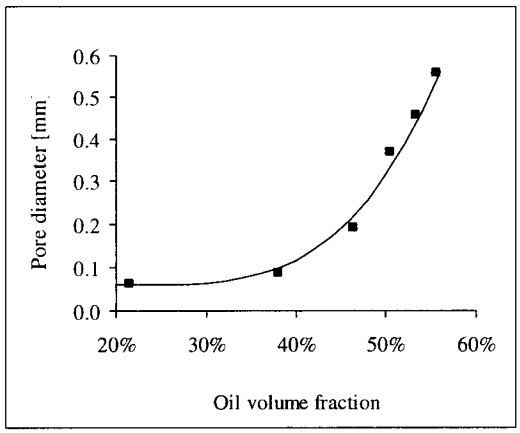


Figure 3: Effect of the oil amount on the pore size of the blocks.

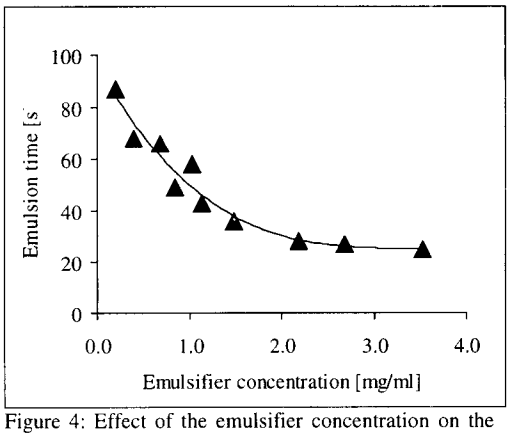


Figure 4: Effect of the emulsifier concentration on the emulsion time of the O/W emulsion

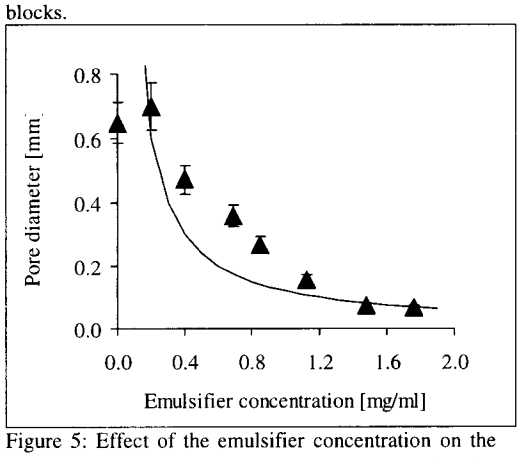


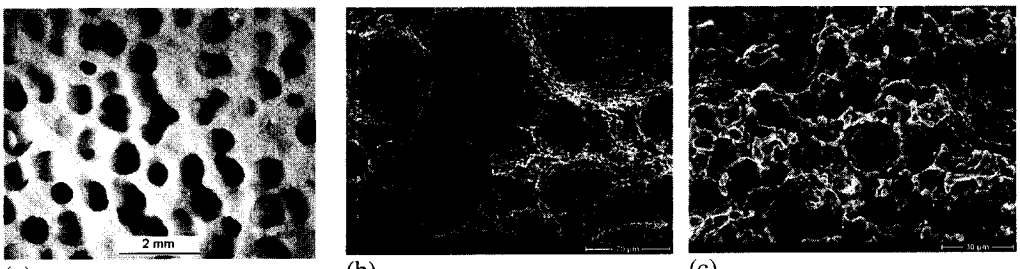
Figure 5: Effect of the emulsifier concentration on the pore size of the blocks. The error bars correspond to the standard errors. The curve has the equation: $y = A/x$ where A is a constant.

The emulsion time was strongly increased by an increase of the oil fraction (**Fig 2**). Above about 58% oil fraction, no O/W emulsion could be obtained, hence leading to an infinite emulsion time (it can only be measured for O/W emulsions). Simultaneously to the increase of emulsion time, there was an increase in pore size (**Fig 3**). The emulsion time was also dependent on the emulsifier amount (**Fig 4**), decreasing from 90s with 0.1 mg/ml PCO to 25s with 3.5 mg/ml PCO. In the latter case, considering the time required to mix all ingredients together, the emulsion formation was practically instantaneous. The addition of larger amount of emulsifier provoked a decrease of pore

size (**Fig 5, Fig 6**). At a high emulsifier concentration, the pores created by the oil had a size in the micrometer range, close to the size of the pores created by the aqueous solution.

DISCUSSION

The occurrence of O/W or W/O emulsions can be explained considering the hydrophilic-lipophilic balance (HLB) of the two emulsifiers. PCO has a value of 12-14 and SM has a value of 4.3 [3]. Paraffin oil requires a HLB value superior to 11 to form a O/W emulsion and inferior to 8 to form a W/O emulsion [4]. Therefore, a O/W emulsion is expected to form when PCO is used as emulsifier and a W/O emulsion is expected to form when SM is used, as observed experimentally. Despite the use of PCO, no O/W emulsion could be obtained over about 60% oil fraction (**Fig1**, **Fig 2**). This is probably due to the fact that the distance between oil droplets became so small that the rate of droplet flocculation became the same as that of droplet formation. A phase inversion is actually expected to occur when the oil fraction exceeds 60% and if no special care is taken [5]. The instability of the emulsion caused by an increase of oil fraction could also explain the increase in pore size (**Fig 3**). When the amount of emulsifier was increased, oil droplets could form more easily, hence leading to a decrease of the emulsion time (**Fig 4**). As the interfacial area O/W is proportional to the amount of emulsifier, the pore size should be inversely proportional to the emulsifier amount. Experimentally, there was a decrease of pore size, but not as fast as predicted by this simple rule (**Fig 5**).



(a) (b) (c)

Figure 6: Microstructure of a block made with 10g paraffin and (a) 0.2 mg/ml PCO emulsifier, (b) 1.5 mg/ml PCO emulsifier and (c) 3.5 mg/ml PCO emulsifier.

CONCLUSION

Calcium phosphate emulsions can be used to make round granules or porous blocks from an injectable paste. The size of the pores can be modified by varying the emulsifier concentration or the oil fraction. The porosity can be modified by varying the oil fraction. The composition of the solid is either a low-temperature phase (e.g. CDHA) or, by sintering the solid, a high-temperature phase (e.g. β-TCP).

ACKNOWLEDGEMENT

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